

current $i = Q/t$ so time $t = Q/i = 4.0/2.0 = 2.0$ s

- 20 A** Combined resistance of R_4 and $R_5 = R/2$
Combined resistance of R_3 , R_4 and $R_5 = 3R/2$
Combined resistance of R_2 , R_3 , R_4 and $R_5 = 3R/5$
By potential divider principle, potential difference across R_1 is greater than potential difference across combined resistance of R_2 , R_3 , R_4 and R_5
Hence, R_1 has the greatest potential difference.

- 21 D** By potential divider principle, voltmeter reading increases when effective resistance